ECE 6950, AQOPT, Fall 2024
Prof. Karan Mehta

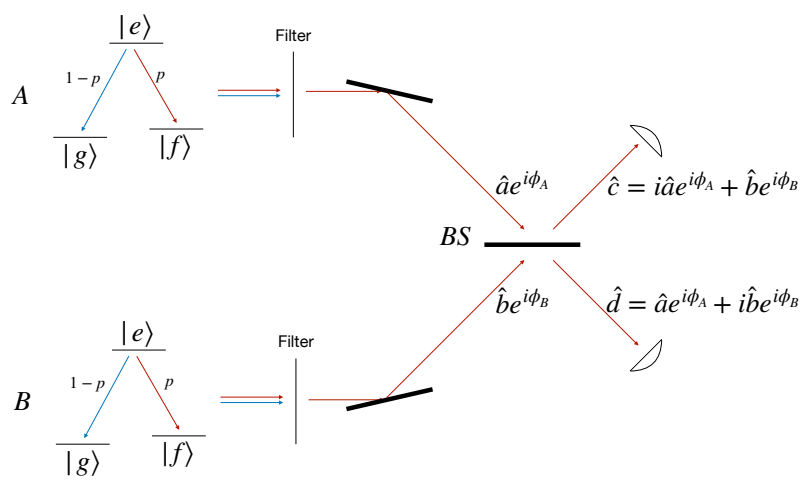
Problem Set 8: Single-photon interference and laser cooling

Assigned: Monday, Nov. 18
Due: Tuesday, Nov. 26

Question 1. Remote entanglement via single-photon detection

This problem considers a scheme for entanglement between distant particles essentially as proposed in L.-M. Duan, M.D. Lukin, J.I. Cirac, and P. Zoller, Nature 414, 413-418 (2001).

Suppose we have two emitters, each of which consists of a 3-level system; we excite both emitters A and B to $|e\rangle$, and allow them to decay with (small) probability $p \ll 1$ to $|f\rangle$ and probability $1-p$ to $|g\rangle$. Each decay is accompanied by emission of a photon with distinct frequencies. We assume we collect these into a single spatial mode with unity efficiency, e.g. employing a high-finesse cavity. We filter out photons of the frequency corresponding to decay to $|g\rangle$. Thus the state of, e.g. atom A and the field in the unfiltered mode after this process is $|\Psi\rangle_A \approx |g,0\rangle_A + \sqrt{p}|f,1\rangle_A$, and likewise for B, with the number representing n for Fock states of the mode transmitted by the filter.



a) Assume we perform identical excitations on each atom, and each atom undergoes the same decay – write the total state of the atom-field system after the filter, and show that

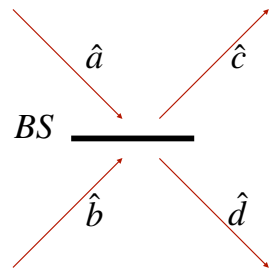
$$|\Psi\rangle_A|\Psi\rangle_B \approx |g,0\rangle_A|g,0\rangle_B + \sqrt{p}(|g,0\rangle_A|f,1\rangle_B + |f,1\rangle_A|g,0\rangle_B) \tag{1}$$

b) The photons travel to a beamsplitter (BS) where they interfere; each path accumulates a phase φ_A or φ_B, so that the field operators at the inputs to the BS are as indicated. Suppose we detect a click on the photodetector at output with field annihilation operator $\hat{c}$, thereby annihilating one photon of this mode; the state of the resulting system can then be written, up to a normalization factor, as $\hat{c}|\Psi\rangle_A|\Psi\rangle_B$. Write the corresponding state, and comment on its features.

c) Given the small probability p assumed here, how many repetitions of the excitation do we have to perform to detect a photon corresponding to creating the state you found in (b)? In practice, over sufficiently long timescales, long optical paths will drift in optical path length, e.g. due to spatiotemporal variations in air pressure or temperature. What do you expect to be the effect of these phase fluctuations on the state found in (b)?

Question 2. Remote entanglement utilizing Hong-Ou-Mandel effect

This problem looks at the use of two-photon interference in the Hong-Ou-Mandel effect to generate entanglement in a more robust fashion, following C. Simon and W. Irvine, PRL 91:11 110405 (2003), and the experimental realization in D. Moehring, et al, Nature 449, 68-71 (2007).



a) Consider a single-mode single photon input to each port of a beamsplitter; i.e, let our wavefunction be represented by $|\Psi\rangle = \hat{a}^\dagger \hat{b}^\dagger |0\rangle$, with $|0\rangle$ representing the vacuum state for all modes. Writing $\hat{a}^\dagger$ and $\hat{b}^\dagger$ in terms of the output field operators, show that this input state is equivalent to a state written in terms of the output modes consisting of a superposition of both photons at one or the other output.

b) Suppose we now have two possible modes at each input, i.e. $\hat{a}_j$ and $\hat{b}_j$ where j can be $\uparrow$ or $\downarrow$; the modes corresponding to $\uparrow$ or $\downarrow$ may be orthogonal polarizations, or different frequencies. Only modes of the same polarization/frequency interfere, so we can write the output mode operators as $\hat{c}_j = \hat{a}_j + i\hat{b}_j$, and $\hat{d}_j = i\hat{a}_j + \hat{b}_j$.

Suppose we can represent our photon wavefunction by one of four possible Bell states, with respect to the input field operators:

$$|\Psi_\pm\rangle_{\rm ph} = \frac{1}{\sqrt{2}}\left(\hat{a}^\dagger_\uparrow \hat{b}^\dagger_\downarrow \pm \hat{a}^\dagger_\downarrow \hat{b}^\dagger_\uparrow\right)|0\rangle \tag{2}$$

$$|\Phi_\pm\rangle_{\rm ph} = \frac{1}{\sqrt{2}}\left(\hat{a}^\dagger_\uparrow \hat{b}^\dagger_\uparrow \pm \hat{a}^\dagger_\downarrow \hat{b}^\dagger_\downarrow\right)|0\rangle \tag{3}$$

Express these states in terms of the output field operators. Assume our detection at ports c and d detects both modes $\uparrow$ and $\downarrow$, and show that only $|\Psi\rangle_-$ can result in simultaneous detection events at both detectors.

c) Now consider two separated atoms, each of which is excited and can decay into one of two ground states $|g_\uparrow\rangle$ and $|g_\downarrow\rangle$ with equal probability, emitting a photon into mode $\uparrow$ or $\downarrow$ respectively, i.e. such that the state of atom A and its emitted photon following decay is $|\Psi\rangle_A = \frac{1}{\sqrt{2}}\left(|g_\downarrow\rangle_A \hat{a}^\dagger_\downarrow + |g_\uparrow\rangle_A \hat{a}^\dagger_\uparrow\right)|0\rangle$. Show that the state of the total atom-photon system $|\Psi\rangle_A|\Psi\rangle_B$ can be written

$$\frac{1}{2}\left(|\Phi_+\rangle_{\rm at}|\Phi_+\rangle_{\rm ph} + |\Phi_-\rangle_{\rm at}|\Phi_-\rangle_{\rm ph} + |\Psi_+\rangle_{\rm at}|\Psi_+\rangle_{\rm ph} + |\Psi_-\rangle_{\rm at}|\Psi_-\rangle_{\rm ph}\right), \tag{4}$$

where

$$|\Psi_\pm\rangle_{\rm at} = \frac{1}{\sqrt{2}}\left(|g_\uparrow g_\downarrow\rangle \pm |g_\downarrow g_\uparrow\rangle\right) \tag{5}$$

$$|\Phi_\pm\rangle_{\rm at} = \frac{1}{\sqrt{2}}\left(|g_\uparrow g_\uparrow\rangle \pm |g_\downarrow g_\downarrow\rangle\right) \tag{6}$$

What is the atomic state conditional upon coincident photon detections at outputs c and d?

d) Now suppose the two optical paths from atoms A and B had a relative phase difference that might vary over time; comment on the effect this has on the photon field and the measurement outcome in this scheme.

Question 3. Radiation pressure and damping forces

This problem calculates radiation pressure and damping forces with a slightly different method which can be useful for intuition, and considers the effect of beam intensity and detuning on the damping forces.

a) Consider a two-level system with states $|g\rangle$ and $|e\rangle$, with a frequency difference ω_0 in a classical plane-wave field $\mathbf{E} = \hat{\epsilon}E_0(e^{i(kz-\omega t)} + \text{c.c.})$, taking E_0 real. The effective mean force can be defined via Ehrenfast's theorem:

$$F = \frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{p}\rangle\,. \tag{7}$$

We can use the basic relation $\frac{\mathrm{d}}{\mathrm{d}t}\left\langle\hat{A}\right\rangle = \frac{i}{\hbar}\left\langle\left[\hat{H},\hat{A}\right]\right\rangle$, and define a (complex) Rabi frequency now including the spatial phase variation of the field: $\Omega(z) = -\frac{dE_0}{\hbar}e^{ikz}$.

Starting from the rotating-frame dipole interaction Hamiltonian in the RWA, show that we can write the force of equation 7 in terms of the density matrix elements of the two-level system as:

$$F = -\hbar\left(\frac{\partial\Omega}{\partial z}\rho_{eg}^* + \frac{\partial\Omega^*}{\partial z}\rho_{eg}\right). \tag{8}$$

b) If changes in position are slow enough with respect to the equilibration timescale ($\sim\Gamma^{-1}$), we can easily calculate this force using steady state solutions for the density matrix elements. With a detuning of a single laser frequency defined as $\delta = \omega - \omega_0$, these steady-state values were (problem set 5):

$$\rho_{eg} = \frac{-i\Omega}{(\Gamma/2 - i\delta)(1+s)}, \tag{9}$$

$$\rho_{ee} = \frac{s_0/2}{1 + s_0 + (2\delta/\Gamma)^2}, \tag{10}$$

where $s_0 = \frac{8|\Omega|^2}{\Gamma^2}$ is the resonance saturation parameter and $s = \frac{s_0}{1+(2\delta/\Gamma)^2}$.

Using these steady state solutions and the result of part (a), show that the force on an atom at rest is

$$F_0 = \frac{\hbar k\Gamma}{2}\frac{s}{1+s}. \tag{11}$$

c) If the atom is not at rest, the frequency seen by the atom is modified by the Doppler shift: $\delta \to \delta - \mathbf{k}\cdot\mathbf{v}$, with $\mathbf{v}$ the atom's velocity. Substituting this detuning into the solution for F_0, and considering situations where the Doppler shift is small (neglecting terms quadratic and above in kv), show that we can write the force as

$$F \approx F_0 - \beta v, \tag{12}$$

where the damping coefficient is given by

$$\beta = -\hbar k^2\frac{4s_0\delta/\Gamma}{(s_0 + 1 + 4\delta^2/\Gamma^2)^2}. \tag{13}$$

What values of s_0 and δ maximize this damping coefficient?

d) Give a physical interpretation of the F_0 in terms of the photon momentum $\hbar k$, remembering that photons are absorbed directionally (from the propagation direction of the beam) and emitted equally likely in opposite directions (dipole emission pattern).

Plot β as a function of rest-frame detuning δ for a few different values of s_0. What values of s_0 and δ maximize the damping? What happens to β and F_0 when we increase the laser intensity for a fixed δ, and how can we explain the difference in behavior?